

Yinxi Li (Larry)

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Education

University of Waterloo, Canada

M.Math. in Computer Science, GPA: 97.7/100, Advisor: [Pengyu Nie](#)

May 2025 – April 2027

(Expected)

The Chinese University of Hong Kong, Hong Kong (SAR)

B.Sc. in Computer Science, [ELITE Stream](#)

Sept 2020 – July 2024

ETH Zürich, Switzerland

D-INFK, Exchange Student

Feb 2023 – Sept 2023

Research Interests

LLM agents, post-training and applications to software engineering, scientific discovery, and mathematics.

Publications and Manuscripts

Conference Papers

Yinxi Li, Yuntian Deng, and Pengyu Nie. “**TokDrift: When LLM Speaks in Subwords but Code Speaks in Grammar.**” In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. [arXiv:2510.14972](#)

Fang Wu*, Weihao Xuan*, Heli Qi*, Hanqun Cao*, Heng-Jui Chang*, Zeqi Zhou*, Haokai Zhao, Ma Jian, Carl Ma, Yu-Chi Cheng, Kuan Pang, Xiangru Tang, Zehong Wang, Guanlue Li, Hanchen Wang, Kejun Ying, Pan Lu, Chiho Im, Seungju Han, Peng Xia, Tinson Xu, **Yinxi Li**, Deyao Zhu, Pheng-Ann Heng, Naoto Yokoya, Masashi Sugiyama, Li Erran Li, Jure Leskovec, Yejin Choi. “**Proteo-R1: Reasoning Foundation Models for De Novo Protein Design.**” In *Proceedings of the International Conference on Machine Learning (ICML)*, 2026. [arXiv:2605.02937](#)

Preprints and Manuscripts Under Review

Weihao Xuan*, Qingcheng Zeng*, Jiarui Liu*, Rui Yang*, Yunze Xiao, **Yinxi Li**, Zeqi Zhou, Heli Qi, Leyang Cui, Tianhao Ma, Hanqun Cao, Junjue Wang, Tianyu Liu, Weihua Du, Yueqi Song, Jiayi Geng, Matthew Yu Heng Wong, Rob Voigt, Chuan Hong, Xiang Yue, Jen-Tse Huang, Xuhai Xu, Yifan Peng, Bernhard Schölkopf, Zhijing Jin, Pheng Ann Heng, Nan Liu, Naoto Yokoya, and Mona Diab. “**Learning from Use: Test-Time Learning in Large Language Models and Agents.**” *Manuscript under review at Nature Reviews Computing.*

Yinxi Li, Jiaao Chen, Fang Wu, Jiakai Yu, Heli Qi, Weihao Xuan, Haokai Zhao, Pengyu Nie, Di Jin, and Xiangru Tang. “**Learning Multi-step Reasoning via Persistent Latent State Propagation.**” In *ICLR 2026 Workshop on Latent & Implicit Thinking – Going Beyond CoT Reasoning*, 2026.

Liliia Hotsko, **Yinxi Li**, Yuntian Deng, and Pengyu Nie. “**Code2LoRA: Hypernetwork-Generated Adapters for Code Language Models under Software Evolution.**” *Manuscript under review at ACL Rolling Review May 2026.* [arXiv:2606.06492](#)

Aaron Tu*, Weihao Xuan*, Heli Qi*, Xu Huang*, Qingcheng Zeng, Shayan Talaei, Yijia Xiao, Peng Xia, Xiangru Tang, Yuchen Zhuang, **Yinxi Li**, Bing Hu, Hanqun Cao, Wenqi Shi, Rui Yang, Nan Liu, Huaxiu Yao, Ge Liu, Li Erran Li, Amin Saberi, Naoto Yokoya, Jure Leskovec, Yejin Choi, and Fang Wu. “**Position: The Hidden Costs and Measurement Gaps of Reinforcement Learning with Verifiable Rewards.**” *Manuscript under review at ACL Rolling Review May 2026.* [arXiv:2509.21882](#)

Fang Wu, Yunxiang MO, Heli Qi, Weihao Xuan, Tinson Xu, Hongbin Lin, Yanchao Li, Zeqi Zhou, Kejun Ying, Hanchen, Peng Xia, Xiangru Tang, Zehong Wang, Ningshan Ma, **Yinxi Li**, Jiakai Yu, James Zou, Jure Leskovec, Yejin Choi. “**Robust Information-Gain Control Active Reasoning under Approximate Beliefs.**” *Manuscript under review at NeurIPS 2026.*

Fang Wu*, Haokai Zhao*, Da Xing, Hanqun Cao, Tinson Xu, Xinyu Xiang, Yanchao Li, Xiangru Tang, Zehong Wang, Kuan Pang, Hanchen, Hongbin Lin, Zeqi Zhou, **Yinxi Li**, Peng Xia, Aaron Tu, Li Erran Li, Molei Tao, Jure Leskovec, Aditya Joshi, Yejin Choi. “**NoiseRater: Meta-Learned Noise Valuation for Diffusion Model Training.**” *Manuscript under review at NeurIPS 2026.* [arXiv:2605.08144](#)

Project Experience

GUI Agent Benchmark for Scientific Tasks

June 2026 – Present

- Building a benchmark for evaluating GUI agents on science-oriented workflows requiring tool use, visual grounding, and multi-step reasoning.

Test-Time Learning for Search Agents on Long-Horizon Retrieval Tasks

April 2026 – Present

- Studying the mismatch between LLM-generated agent queries and dense retriever embedding spaces in complex dense retrieval settings.
- Developing test-time adaptation strategies that use retrieval feedback to refine agent search behavior across multi-step information-seeking tasks.

Self-Evolving LLM Agents for Machine Learning Engineering

March 2026 – Present

- Designing a Git-managed self-evolving harness that records agent actions, code changes, experimental outcomes, and reusable feedback for future task attempts.

Unit Test Transplantation with Retrieval-Augmented Generation

Sept 2024 – Dec 2024

- Developed a RAG pipeline for test transplantation to improve unit test generation in API frameworks.
- Analyzed model performance on 2,000+ unit test cases extracted from 7 diverse open-source repositories, identifying optimization opportunities and proposing enhancement.

Optimizing Structure of Vision-Language Retrieval System

Aug 2023 – June 2024

ELITE Stream Final Year Project, Supervisor: [Eric Lo](#) 

- Developed and integrated a proxy model and optimized retrieval structures, achieving up to 50% faster object detection and 1000× inference speedup with minimal accuracy loss.
- Designed a vision-language retrieval system for large-scale video datasets, enabling efficient and diverse top-K results with reduced computational complexity from $\mathcal{O}(n^2d)$ to $\mathcal{O}(nd)$.
- Conducted comprehensive evaluations, including baseline comparisons and ablation studies, demonstrating superior retrieval accuracy, diversity, and architectural effectiveness.

Adaptive Streaming for Short-Video Services

June 2022 – Aug 2022

CUHK Summer Research, Supervisor: [Lee Yiu-Bun](#) 

- Implemented and evaluated multiple adaptive bitrate (ABR) algorithms (including BBA, BOLA, and the RL-based Pensieve), and tuned the simulator to support focused testing on short video samples.

Competition

World University Students' Digital Intelligence Competition Invitational Tournament (by Tencent)

April 2023 - July 2023

Ranked first among overseas institutions, Mentor: [Tan LEE](#) 

- Developed a multi-agent reinforcement learning framework for Honor of Kings, with optimized reward functions and customized agent strategies.

Research Internship Experience

Research Assistant

Remote

Southern University of Science and Technology, Supervisor: [SONG Xuan](#) 

Aug 2024 - Oct 2024

- Developed an expert knowledge collaboration system for the automotive quality inspection value chain based on large language model and retrieval-augmented generation.

Research Intern

Shenzhen, China

Research Institute of Tsinghua University in Shenzhen

April 2024 - June 2024

- Implemented statistical, ML, and deep learning methods for multivariate time-series anomaly detection.
- Utilized a variational autoencoder for anomaly detection, improving customer load monitoring.

Research Intern

Changchun, China

Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences

June 2020 - Aug 2020

- Optimized object detection of moving objects in specific directions in videos.

Research Community Service

Teaching Assistant: CS 246, CS346, CS446, CS486 (25 Spring, Fall, 26 Winter, Spring, UWaterloo)

Sub-Reviews: ASE 2025, ARR Oct 2025

Association Activities Experience

CUHK Mainland Undergraduate Association (19th & 20th cabinet)

Aug 2021 - July 2023

- Principal of Information Department: Managed communication channels
- Assistant of Sports Department: Coordinated Sports Festival and e-sports tournaments

CUHK Mainland Undergraduate Association Orientation Camp

Aug 2021 - July 2023

- Group Leader(2021-23): Mentored 7 groups and supported 60+ freshmen in academic and campus life

Additional Information

Open-Source Codebase Experience: Hands-on experience using, extending, and modifying LLM/agent inference, model-training and informational retrieval codebases, including vLLM, SGLang, LangChain, LangGraph, Hugging Face Transformers, Accelerate, Tokenizers, Megatron-LM, verl, FAISS, and Tevatron.

Selected Coursework: Machine Learning for SE, Computational Linguistics, Mining & Analysis of SE Data, Foundations of Optimization, Machine Learning Theory, Fundamentals of AI, Rigorous Software Engineering, Introduction to ML, Data Modelling and Databases

Programming Languages: Python, Java, Shell Script